Find the Mistake: Composition Order and Inverse Domains Answer Key

Algebra 2

Quick Answers

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1. f(g(2)) = 17, g(f(2)) = 27
2. the corrected value = 8, the diagnosis = —
3. (a) = 21, (b) = $\frac{x-12}{3}$
4. (a) = 3, (b) = —
5. (a) = $x^2 - 8x + 16$, (b) = 9, (c) = —
6. (a) = $(x-3)^2$, (b) = —
7. (a) = $\frac{2x+5}{x-1}$, (b) = $\frac{11}{2}$, (c) = —
8. (a) = $x^2 + 6x + 9$, (b) = -1, (c) = —

What is verified

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13 of 19 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 2, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2
HSF-BF.A.1c – *problems 1, 2, 5, 8*
HSF-BF.B.4 – *problems 3, 4, 6, 7*
Difficulty 1–5 of 5 across 19 tagged checks.

Common wrong answers

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2. *If they got* -2*: applied the two rules in the reverse order, computing g(f(6)) instead of f(g(6))*
4. *If they got* -3*: assumed the square root always returns the original input, so the round trip gives -3 back*
5. *If they got 45: squared the input first and subtracted 4 afterwards, which is the other composition*
7. *If they got 0.125: flipped the numerator and denominator of the original rule instead of solving for the input*

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1. $f(x) = 3x - 1$, $g(x) = x^2 + 2$. (a) $f(g(2))$. (b) $g(f(2))$.

Solution (a): Inner rule first: $g(2) = 2^2 + 2 = 6$, then $f(6) = 3(6) - 1$. As a single rule, $f(g(x)) = 3(x^2 + 2) - 1 = 3x^2 + 5$. $\boxed{17}$

Solution (b): Now f is the inner rule: $f(2) = 3(2) - 1 = 5$, then $g(5) = 5^2 + 2$. As a single rule, $g(f(x)) = (3x - 1)^2 + 2$. $\boxed{27}$

The point of the pair: same two rules, same input, two different answers. Composition is not commutative, and this is the fact every later problem leans on.

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2. A student gave -2 for $(f \circ g)(6)$ with $f(x) = \frac{x}{2} + 7$, $g(x) = 8 - x$. (a) Correct value. (b) Which composition did the student do?

Solution (a): The inner rule is g, so it acts on 6 first.

$$g(6) = 8 - 6 = 2$$
$$f(2) = \frac{2}{2} + 7 = 1 + 7$$

So the correct value is $\boxed{8}$

Solution (b) — instructor-judged. The student computed $g(f(6))$: $f(6) = 3+7 = 10$, then $g(10) = 8 - 10 = -2$. A correct response says the rules were applied in the wrong order — g must act on the input first, f second — or equivalently identifies -2 as $g(f(6))$. Naming which rule was used first is required; "arithmetic slip" is not a diagnosis and earns nothing.

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3. $f(x) = 3x + 12$; a student claims the inverse is $\frac{x}{3} + 12$. (a) Round-trip 5. (b) Correct $f^{-1}(x)$.

Solution (a): $f(5) = 3(5) + 12 = 27$. Feeding 27 into the claimed rule:

$$\frac{27}{3} + 12 = 9 + 12$$

The round trip returns $\boxed{21}$

— not the 5 it started from, so the claim is false.

Solution (b): Write $x = 3y + 12$ and solve for y.

$$x - 12 = 3y$$
$$y = \frac{x - 12}{3}$$

The 12 must be removed *before* dividing, because it was added last. $\boxed{f^{-1}(x) = \frac{x - 12}{3}}$

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4. A student claims $\sqrt{x-1}$ inverts $h(x) = x^2 + 1$ for every input. (a) Round-trip -3. (b) What does that show?

Solution (a): $h(-3) = 9 + 1 = 10$, then $\sqrt{10 - 1} = \sqrt{9}$. The output is $\boxed{3}$

Solution (b) — instructor-judged. The round trip returns 3, not the -3 it started from, so the claimed rule is not an inverse on all inputs. The reason: squaring destroys the sign, so -3 and 3 share the output 10, and no single rule can send 10 back to both. Full credit needs the failed round trip *and* the two-inputs-one-output (not one-to-one) reason. A student who adds that the claim becomes true once the inputs of h are restricted to $x \geq 0$ has the complete picture.

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5. A student wrote $(g \circ f)(x) = x^2 - 4$ for $f(x) = x - 4$, $g(x) = x^2$. (a) Correct rule. (b) Its value at 7. (c) The error.

Solution (a): f is the inner rule, so subtract first and square the result.

$$\begin{aligned}(g \circ f)(x) &= g(x-4) = (x-4)^2 \\ &= x^2 - 8x + 16\end{aligned}$$

$$\boxed{(g \circ f)(x) = x^2 - 8x + 16}$$

Solution (b): $(7-4)^2 = 3^2$, giving $\boxed{9}$

Solution (c) — instructor-judged. The student squared first and subtracted afterwards, which is $(f \circ g)(x) = x^2 - 4$ — the other composition. At $x = 7$ that rule gives 45 instead of 9. A correct response names the *order*: the inner rule f subtracts, and only then does g square. Reporting the two different numbers without naming the order is not a diagnosis.

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6. $k(x) = \sqrt{x} + 3$. (a) Find $k^{-1}(x)$. (b) State and justify the restriction it must carry.

Solution (a): Write $x = \sqrt{y} + 3$ and undo the $+3$ before squaring.

$$\begin{aligned}x - 3 &= \sqrt{y} \\ y &= (x-3)^2\end{aligned}$$

$$\boxed{k^{-1}(x) = (x-3)^2}$$

Solution (b) — instructor-judged. Required: inputs of the inverse must satisfy $x \geq 3$. The justification: $\sqrt{x}$ is never negative, so k never outputs anything below 3; the inverse's inputs are exactly the outputs of k. Without the restriction the squaring rule would happily accept $x = 1$ and return 4, but $k(4) = 5$, not 1 — the round trip fails. Full credit needs the restriction *and* its tie to the range of k; the counterexample is the strongest form.

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7. A student claims the inverse of $f(x) = \frac{x+5}{x-2}$ is $\frac{x-2}{x+5}$. (a) Correct $f^{-1}(x)$. (b) Its value at 3. (c) Why upside down is wrong.

Solution (a): Write $x = \dfrac{y+5}{y-2}$, clear the fraction, collect y.

$$\begin{aligned}x(y-2) &= y + 5 \\ xy - 2x &= y + 5 \\ xy - y &= 2x + 5 \\ y(x-1) &= 2x + 5\end{aligned}$$

$$\boxed{f^{-1}(x) = \frac{2x+5}{x-1}}$$

Solution (b): $\dfrac{2(3)+5}{3-1} = \dfrac{11}{2}$, so $\boxed{f^{-1}(3) = \dfrac{11}{2}}$

The student's upside-down rule gives $\frac{3-2}{3+5} = \frac{1}{8}$ instead — nowhere near.

Solution (c) — instructor-judged. Turning a rule upside down produces its *reciprocal*, $1/f(x)$, which undoes multiplication only. An inverse undoes the entire rule and is found by solving $x = f(y)$ for y. A correct response draws that distinction, or demonstrates it: $f\left(\frac{x-2}{x+5}\right)$ does not simplify to x.

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8. $f(x) = x + 3$, $g(x) = x^2$; $(f \circ g)(x) = x^2 + 3$. (a) $(g \circ f)(x)$. (b) The one input where they agree. (c) Why only one.

Solution (a): $g(f(x)) = (x + 3)^2$, expanded:

$$(x + 3)^2 = x^2 + 6x + 9$$

$$\boxed{(g \circ f)(x) = x^2 + 6x + 9}$$

Solution (b): Set the two composed rules equal.

$$\begin{aligned} x^2 + 3 &= x^2 + 6x + 9 \\ 0 &= 6x + 6 \\ x &= -1 \end{aligned}$$

Check: $(-1)^2 + 3 = 4$ and $(-1 + 3)^2 = 4$ ✓ $\boxed{x = -1}$

Solution (c) — instructor-judged. When the two rules are set equal the x^2 terms cancel, leaving the linear equation $6x + 6 = 0$, and a linear equation with a nonzero slope has exactly one solution. A correct response names that cancellation (or argues that the difference of the two rules, $6x + 6$, is linear and not constant), so the graphs cross once. Asserting only "they are different functions" does not answer the question.
